Mayur Sai Yaram
AI/ML and Data Engineer
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PROFESSIONAL SUMMARY:

- Data Engineer and AI/ML Engineer with 6+ years of experience in building scalable data infrastructure and developing production-ready AI models.
- Skilled in designing automated data pipelines, developing machine learning solutions, and operationalizing AI using modern cloud, big data, and MLOps tools.
- Adept at bridging the gap between raw data and intelligent decision-making by collaborating with cross-functional teams across engineering, analytics, and product domains.
- Proficient in building end-to-end pipelines for data ingestion, transformation, feature engineering, model training, deployment, and monitoring across cloud and on-prem environments.
- Hands-on experience with big data ecosystems (Spark, Kafka, Hadoop), cloud services (AWS, GCP, Azure), and MLOps tools (MLflow, Kubeflow, SageMaker, Docker, Kubernetes).
- Proven track record of deploying ML models (classification, regression, forecasting, NLP) into production environments and optimizing them using hyperparameter tuning and model explainability techniques (SHAP, LIME).
- Expert in translating business problems into data-driven solutions using a combination of data engineering, statistical modeling, and machine learning.
- Experience integrating LLMs and generative AI solutions into enterprise systems using OpenAI, LangChain, vector databases (FAISS, Pinecone), and RAG pipelines.
- Adept at creating automated CI/CD pipelines for ML models and ETL jobs using Git, Jenkins, and Airflow, reducing manual effort and increasing model retraining speed.
- Strong background in relational and NoSQL databases (PostgreSQL, Snowflake, MongoDB), combined with deep knowledge of SQL optimization and data quality frameworks.
- Excellent communicator with a collaborative mindset, working cross-functionally with product managers, analysts, and data scientists to deliver scalable, high-impact AI solutions.
- Versatile Data and AI/ML Engineer with extensive experience in building reliable data platforms and deploying scalable machine learning systems in cloud-native environments.
- Specialized in creating robust ETL/ELT workflows and architecting ML pipelines that transform raw data into actionable intelligence for business-critical use cases.
- Deep expertise in feature engineering, model lifecycle management, and real-time model inference using tools like Spark, Airflow, MLflow, and TensorFlow.
- Strong knowledge of large-scale data processing and model deployment on platforms like AWS SageMaker, GCP Vertex AI, and Azure ML.
- Led the integration of generative AI (LLMs) into internal tools, enhancing search, document summarization, and customer service automation.
- Passionate about bridging the gap between data engineering and machine learning by building unified, production-grade data and ML platforms.

TECHNICAL SKILLS:

Data Engineering	Apache Spark, Kafka, Airflow, Hadoop, dbt, Delta Lake
Databases	Snowflake, PostgreSQL, MySQL, MongoDB, BigQuery
ML & AI Tools	Scikit-learn, TensorFlow, PyTorch, XGBoost, Hugging Face
MLOps & DevOps	MLflow, Docker, Kubernetes, Git, CI/CD, Feast
Cloud Platforms	AWS (S3, Glue, EMR, SageMaker), GCP (Vertex AI, BigQuery), Azure
Languages	Python, SQL, Scala, Bash
Visualization & Analytics	Power BI, Tableau, matplotlib, Plotly
Frameworks	FastAPI, LangChain, Flask

PROFESSIONAL EXPERIENCE:

UBS - Weehawken, New Jersey **Jan 2024 - Present**
AI/ML & Data Engineer

Responsibilities:
- Designed and implemented scalable data pipelines using Apache Spark, Airflow, and Kafka to support high-volume ML workloads.
- Built robust ETL/ELT workflows to ingest and process data from APIs, databases, and event streams into cloud data lakes (S3, GCS).
- Implemented data quality checks and pipeline monitoring using Great Expectations, custom alerts, and Airflow SLA configurations.
- Automated CDC (Change Data Capture) processes using Debezium and Kafka Connect for near real-time updates to ML pipelines.
- Deployed infrastructure-as-code (IaC) for data pipelines using Terraform to ensure reproducibility and version control.

- Built and deployed machine learning models (XGBoost, Random Forest, Neural Networks) for use cases like churn prediction, fraud detection, and lead scoring.
- Developed deep learning models for NLP tasks (text classification, summarization, sentiment analysis) using Hugging Face Transformers and TensorFlow.
- Deployed ML models as REST APIs using FastAPI + Docker and managed them using Kubernetes for autoscaling and resiliency.
- Created automated retraining pipelines triggered by data drift or model performance degradation using scheduled jobs in Airflow.
- Designed and implemented a centralized feature store using Feast to streamline reuse of ML features across teams and projects.
- Set up model monitoring dashboards to track inference latency, drift, and performance in production using Prometheus and Grafana.
- Partnered with product managers, analysts, and stakeholders to identify high-impact ML use cases and deliver production-grade solutions.
- Contributed to the design of internal ML platforms that standardized data preprocessing, model deployment, and performance tracking.

Environment: Apache Spark, Apache Airflow, Apache Kafka, Kafka Connect, Debezium, dbt, AWS (S3, Glue, Lambda, SageMaker), Google Cloud Platform (GCS, BigQuery), Azure, Amazon S3, Google Cloud Storage, PostgreSQL, Snowflake, MongoDB.

Capgemini - Bengaluru, India **Apr 2021 - July 2023**
AI/ML Engineer

Responsibilities:
- Built and deployed machine learning models that forecasted customer activity, providing live data prediction and network traffic optimization.
- Analyzed massive data, performed feature engineering, and turned unstructured data into structured information for data models to get better insights and accurate predictions.
- Developed ML algorithms using Python, TensorFlow, and Scikit-learn to implement the code for various models and deployed them on AWS.
- Advised and debugged model performance; retrained models based on new data to ensure high accuracy and efficiency.
- Designed and implemented RAG-based AI solutions for question-answering systems, enabling the generation of contextually accurate and up-to-date responses.
- Reported findings and results to stakeholders using Tableau, providing actionable recommendations that would help these business outcomes.
- Developed GenAI-based content personalization algorithms, providing tailored recommendations and content based on customer interaction history and preferences.
- Managed end-to-end model training pipelines by utilizing SageMaker's built-in algorithms and custom containers to streamline the process.
- Led the integration of SageMaker with AWS data sources such as S3 and Redshift for automated data ingestion and pre-processing workflows.
- Integrated OpenAI APIs for building conversational AI solutions, including chatbots and virtual assistants, to improve customer engagement and support.
- Led experiments to optimize generative models for specific tasks, including image-to-text synthesis and text summarization, improving quality and coherence.
- Designed and implemented data pipelines to transfer large-scale AI/ML training datasets to Cosmos DB for real-time analytics.
- Collaborated with cross-functional teams to integrate generative AI models into production systems, enabling real-time content generation for various applications.
- Created Splunk dashboards to monitor real-time metrics for AI models, providing stakeholders with insights into model predictions and accuracy.
- Automated data processing and model training with repetitive tasks, to assist in workflow improvements as well as efficiency enhancements.
- Evaluated the performance of RAG models using metrics like precision and recall, and iteratively improved model accuracy.
- Optimized workflow automation using Langchain to streamline data processing tasks, enhancing efficiency in data retrieval and decision-making processes.
- Automated the CI/CD pipeline for machine learning models through OpenShift, improving deployment efficiency and reducing time to production.
- Integrated machine learning models into the software applications and services so that programs can work seamlessly without disturbing user experiences.

Environment: Python, R, SQL, TensorFlow, Keras, Scikit-learn, Hadoop, OpenShift, Langchain, Cosmos DB, Sagemaker, Spark, GenAI, MLFlow, NLTK, Tableau, Git, GitHub, Splunk, OpenSearch, Agile Scrum, JIRA, Confluence

Accenture - Hyderabad, India **Aug 2020 - Apr 2021**
Data Engineer

Responsibilities

- Participated in Data Acquisition with the Data Engineering team to extract clinical and imaging data from several data sources like flat files and other databases.
- Developed Spark scripts using Python on Azure HDInsight for Data Aggregation, Validation and verified its performance over MR jobs.
- Performed Data Visualization and Designed Dashboards with Tableau and generated complex reports including charts, summaries, and graphs to interpret the findings for the team and stakeholders.
- Wrote documentation for each report including purpose, data source, column mapping, transformation, and user group.
- Extracted and loaded data into the Data Lake environment (MS Azure) by using Sqoop which was accessed by business users.
- Developed Python programs for manipulating the data reading from various Teradata and converting them as one CSV Files.
- Architect & implement medium to large scale BI solutions on Azure using Azure Data Platform services (Azure Data Lake, Data Factory, Data Lake Analytics, Stream Analytics, Azure SQL DW, HDInsight/ Databricks, NoSQL DB).
- Worked on migration of data from On-prem SQL server to Cloud databases (Azure Synapse Analytics (DW) & Azure SQL DB).
- Responsible for building scalable distributed data solutions using Azure Data lake, Azure Databricks. Azure HDInsight & Azure Cosmos DB.
- Designed and implemented solutions using Microsoft Customer Data Platform (CDP) to manage and analyze customer data.
- Collaborated with cross-functional teams to gather business requirements and translated them into technical specifications.
- Experience Configuring and managing AzureAD Connect, AzureAD Connect health, Microsoft Azure Active Directory.
- Integrated the CDP with various data sources, ensuring seamless data flow and accuracy while developing and maintaining efficient data pipelines.
- Experience working with Azure BLOB and Data Lake storage and loading data into Azure SQL Synapse analytics (DW).
- Developing ETL data workflows and code for new data integration and maintain/enhance existing workflows and code.

Environment: Snowflake, Datalake, Lakehouse, Hive, pyspark, Git, Jenkins, Kafka, Jira, Qlick View, Qlik Sense, SQL, Scala, JUnit, MySQL, Power BI, Databricks, Python, Azure, Azure Data Lake, Adobe Analytics, Docker.

Harman - Hyderabad, India **May 2019 - July 2020**
Data Engineer

Responsibilities:

- Migrate data (Amazon Connect Call Metrics) from legacy systems (Cassandra) to cloud based (AWS) Snowflake Data Warehouse
- As a Data Engineer, assisted in leading the plan, building, and running states within the Enterprise Analytics Team.
- Design and Develop ETL Processes in AWS Glue to migrate Campaign data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift
- Proactively identify new opportunities to support the business with data, aggregate various data sets to inform business decisions
- Migrate data from legacy systems (Salesforce) to cloud based (AWS) Snowflake Data Warehouse and develop metrics to provide data insights as per business requirement
- Developed PySpark based pipelines using spark data frame operations to load data to EDL using EMR for jobs execution & AWS S3 as storage layer.
- Iterate rapidly and work collaboratively with product owners, developers, and other members of the development team.
- Developed and optimized OBIEE RPD models, creating complex metadata layers to enhance data modeling and report generation for semiconductor production insights.
- Integrated AWS SNS and SQS for event-driven messaging, ensuring real-time alerts for anomalies in semiconductor wafer production, and configured AWS Route 53 for efficient network routing.
- Leveraged Alteryx Designer and Alteryx SDK to automate ETL workflows, enabling efficient data preparation, transformation, and blending from multiple sources for semiconductor manufacturing analytics.
- Worked with Cloudera Hadoop distribution, utilizing Kafka for real-time data ingestion and processing, and Ambari for cluster monitoring and management.
- Defect tracking and resolving bug-fixes, hot-fixes by merging the feature branch code to the master branch in GIT source control management.
- Use Amazon Elastic Cloud Compute (EC2) infrastructure for computational tasks and Simple Storage Service (S3) as Storage mechanism.
- Spin up an EMR cluster in AWS to run PySpark applications and SSH to master instance to perform spark-submit options to run application in Client and Cluster mode.
- Configured alerting rules and setup PagerDuty alerting for Elasticsearch, Kafka, Logstash and different Microservices in Kibana
- Complied interactive dashboards in Tableau Desktop and published them to Tableau server which allowed Story telling with Quick filters for on demand needed information and data insights with just one click of a button.

- Schedule AWS snowflake reports through Python airflow, data visualization in AWS S3 cloud box. Automatically upload S3 cloud box files to databases in AWS snowflake through Python airflow.
- Create and maintain source codes in GitHub, kept track of source codes, explored and shared the changes of coding scripts, notes in GitHub.

Environment: AWS, AWS S3, redshift, Bedrock, EMR, Lambda, SNS, SQS, athena, glue, Alteryx Design, Alteryx SDK, cloudwatch, kinesis, route53, IAM, OOZIE, airflow, Teradata, oracle, PowerBI, SQL.

EDUCATION:

University of Florida – Gainesville, Florida **Aug 2023 – May 2025**
Master's in computer science.

Gandhi Institute of Technology and Management – Vizag, India **July 2016 – June 2020**
Bachelor of Technology in Computer Science & Engineering.